NRES 746, Week 12, Lecture 1: Formula to Model (take-home)

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

Due Monday, Nov. 16. Work on paper; no code is needed. You may use the Week 12 web page and your notes.

Part 1: Counting parameters

You measure the height of sagebrush seedlings in quadrats. Quadrats are grouped into plots, and plots into sites; each quadrat has a shrub cover value (`cover`) and each site an elevation (`elev`). Different observers measured different quadrats.

1a. For each mixed model, count the fixed effects and the variance and correlation parameters (include the residual SD σ in that column), then the total.

Model formula	Fixed effects	Variance and correlation	Total
`height ~ cover + (1 | site)`	*2*	*2*	*4*
`height ~ cover + (cover | site)`	*2*	*4*	*6*
`height ~ cover + (cover || site)`	*2*	*3*	*5*
`height ~ cover + (1 | site/plot)`	*2*	*3*	*5*
`height ~ cover + elev + (cover + elev | site)`	*3*	*7*	*10*
`height ~ cover + (1 | site) + (1 | observer)`	*2*	*3*	*5*

1b. Which of the six models has nested random effects, and which has crossed random effects? What would make `site` and `observer` crossed rather than nested?

> Nested: `(1 | site/plot)` (each plot is in one site). Crossed: `(1 | site) + (1 | observer)`, if observers work at more than one site and each site is measured by more than one observer. If each observer worked at only one site, observer would be nested in site.

Part 2: From formula to equations

2a. Write the second model, `height ~ cover + (cover | site)`, as equations: the mean of an observation, the distribution of the residual, and the distribution of each site's random effects. Name every parameter you counted in 1a.

> For quadrat i at site j:
>
> $$\text{height}_{ij} = (\beta_0 + a_j) + (\beta_1 + b_j)\,\text{cover}_{ij} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \text{Normal}(0, \sigma^2)$$
>
> $$\begin{pmatrix} a_j \\ b_j \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_a^2 & \rho\,\tau_a\tau_b \\ \rho\,\tau_a\tau_b & \tau_b^2 \end{pmatrix} \right)$$
>
> Parameters: β_0, β_1 (fixed); τ_a, τ_b, ρ, σ (variance and correlation). Total 6.

2b. For the nested model, `height ~ cover + (1 | site/plot)`, call the among-site SD τ_s, the among-plot (within-site) SD τ_p, and the residual SD σ. Fill in the covariance between the residual heights of two quadrats that are:

- in the same plot: $\tau_s^2 + \tau_p^2$
- in different plots at the same site: τ_s^2
- at different sites: 0

Part 3: Random slopes change the variance

Return to the drought-sensitivity model on the Week 12 web page:

$$\text{width} = (\mu_a + a_j) + (\mu_b + b_j)\,x + \varepsilon, \qquad (a_j, b_j) \sim \text{MVN}(0, G), \qquad \varepsilon \sim \text{Normal}(0, \sigma^2)$$

where x is precipitation in 10-cm units centered at 20 cm. Use $\tau_a = 0.15$, $\tau_b = 0.10$, $\rho = -0.6$, and $\sigma = 0.12$. Recall $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$ and $\text{Var}(cX) = c^2\,\text{Var}(X)$.

3a. Find the variance of one observation's width, for a stand you haven't sampled, as a function of x. Then evaluate it at $x = -1$ (a 10-cm winter) and $x = 1$ (a 30-cm winter).

> $$\text{Var(width)} = \tau_a^2 + 2\rho\,\tau_a\tau_b\,x + \tau_b^2 x^2 + \sigma^2$$
>
> At $x = -1$: $0.0225 + 0.018 + 0.01 + 0.0144 = 0.0649$ mm^2 (SD 0.255 mm).
>
> At $x = 1$: $0.0225 - 0.018 + 0.01 + 0.0144 = 0.0289$ mm^2 (SD 0.170 mm).

3b. Ignore σ and look only at the spread among stands' expected widths, $\text{Var}(a_j + b_j x)$. At what precipitation (in cm) is this spread smallest? What does that mean for the stands' precipitation response lines?

> $\frac{d}{dx}\left(\tau_a^2 + 2\rho\tau_a\tau_b x + \tau_b^2 x^2\right) = 2\rho\tau_a\tau_b + 2\tau_b^2 x = 0$, so $x = -\rho\tau_a/\tau_b = 0.9$, or 29 cm. The minimum variance is $\tau_a^2(1 - \rho^2) = 0.0144$ (SD 0.12 mm).
>
> The stands' lines are closest together near 29 cm and fan out on either side, most in dry winters.

3c. In a sentence or two: what does $\rho = -0.6$ imply about which stands differ most in dry years?

> Stands with low average growth have steeper slopes, so in dry years they fall furthest below the others: differences among stands are largest in drought, and the slow-growing stands are the ones hit hardest.

Part 4: What the sampler sees

4a. The Stan program on the Week 12 web page fits the same model with $J = 12$ stands. Count the unknown quantities Stan samples (everything in the `parameters` block; for `L`, count only the values that are free to vary).

How many does `lmer()` estimate for the same model?

> Stan: `mu` 2 + `tau` 2 + `L` 1 + `z` $2 \times 12 = 24$ + `sigma` 1 = 30. `lmer()`: 6 (μ_a, μ_b, τ_a, τ_b, ρ, σ).

4b. Why the difference? How would each count change with 50 stands?

> `lmer()` integrates the stand effects out of the likelihood; Stan samples them as parameters. With 50 stands: Stan 106, `lmer()` still 6.